MORPHASE Quick Start Guide

MORPHASE is firmware that turns the M5Stack **Cardputer ADV** into a **4-operator FM polyphonic synthesizer**. You play it from a MIDI keyboard, and the Cardputer becomes the editor and controller for the sound. You can also move the sound by tilting the unit or waving your hand over a sensor.

This guide takes you from power-on to your first sound and a quick tour of the main features. For a full description of every setting, see the Reference Guide.

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1. What you need

Item	Details
Cardputer ADV	The original Cardputer (non-ADV) is not supported
microSD card	FAT32. Patches and settings are saved here
MIDI keyboard	Connected in one of these ways
└ Standard	Plug an M5Stack **Unit MIDI** (U187) into the unit's Grove port and connect the keyboard with a 5-pin MIDI cable
└ BLE	An app or keyboard that sends Bluetooth MIDI, on iPhone / iPad etc. (Unit MIDI works too). Tested app: KeyPad
Earphones	Recommended. The built-in speaker barely reproduces low notes

MORPHASE comes in two builds, **Standard** and **BLE**. They differ in whether they accept Bluetooth MIDI and in how many notes can sound at once.

	Standard	BLE
MIDI	Unit MIDI (DIN)	Unit MIDI (DIN) + Bluetooth MIDI
Notes at once	12	10

The startup screen and `SET > Info` show which build you have (the BLE build adds `BLE`).

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2. Start it up

Insert the microSD card and turn the power on.

The first time, a `/MORPHASE` folder is created on the card and **32 factory patches** (ready-made sounds) and 2 factory chains (explained later) are written to it. This takes a few seconds.

After the startup screen, the **PLAY screen** appears. This is MORPHASE's "home screen".



Area	Shows
Top row	Patch name (large), its position in the list (e.g. `1/32`), the algorithm and the number of notes sounding (e.g. `Algo 5 0/12`)
Row below	The notes sounding (one box per voice; with ARP On, the ARP's chord)
Center left	The output waveform (scope). Above it: GMorph and Feedback values, the FX letters (`C D R X V`), `GL`, the chain state and more
Bottom left	Pitch bend bar, ARP and LFO state, the `MIDI` `HOLD` `OCT` `TRN` `THR` indicators
Right column	Tilt (IMU) pad and X / Y state, tilt destinations, master volume (`VOL`)

The tabs `PLAY` `OP` `ALGO` `ARP` `LFO` `FX` `SET` run across the top of the screen.

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3. Connect

Standard build (Unit MIDI)

1. Set the Unit MIDI DIP switch to **BYPASS** and the unit's Grove 5V switch to **5VOUT**
2. Plug the Unit MIDI into the unit's Grove port (G1 / G2) and connect it to your keyboard's MIDI OUT
3. Play a key. If `MIDI` lights at the bottom left of the PLAY screen, MIDI is coming in

The receive channel starts as **Omni** (all channels).

BLE build (Bluetooth MIDI)

1. When powered on, the unit waits for a Bluetooth MIDI connection under the name `MORPHASE`. A dim `b` appears right of the patch name on the PLAY screen
2. On iPhone / iPad, choose `MORPHASE` **inside the app** (its Bluetooth MIDI connection screen), not in the phone's Bluetooth settings
3. Once connected, it changes to a bright `B`



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4. Play your first sound

Play a few keys. The first sound is `01_EPiano` (electric piano). The harder you play, the brighter and louder it gets.

Choosing a patch (sound)

- **On the unit**: press `Tab` a few times to reach the **SET** tab → `Enter` on `Patch` → `Enter` on `Load` → choose with `;` `.` and press `Enter`
- **From MIDI**: Program Change 0–31 from your keyboard selects factory patches 01–32

Five to try first

Patch	What it sounds like
`01_EPiano`	Electric piano. Its character follows how hard you play
`04_Organ`	Organ. Sounds as long as you hold the keys
`13_MorphLd`	Mono lead. Tilt the unit to change the tone (next chapter)
`23_TiltPad`	Wide pad. Tilt to move the tone
`25_Kick`	Kick drum. Play low keys around C1–C2

Basic keys

Key	Action
`K` / `L`	Volume down / up (hold to keep changing)
`;` / `.`	Octave up / down (PLAY screen)
`,` / `/`	Transpose in semitones (PLAY screen)
`Z` / `X`	Pitch bend down / up while held
`D`	Note hold (keeps the notes you hold sounding until you press it again)
`Shift+H`	Show / hide the help for the current screen

`G0` (the unit's G0 button)	Jump between the PLAY screen and the last editing tab you used



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5. Tilt the unit

The Cardputer ADV has a tilt sensor (IMU). **It starts Off**, so turn it on first.

Key	Action
`Shift+A`	X axis (left/right tilt) On / Off
`Shift+S`	Y axis (front/back tilt) On / Off
`A` / `S`	Freeze (hold) / release the current tilt of that axis

Once it's on, tilting changes **the tone (waveform) on the X axis and the grit (feedback) on the Y axis** in most patches. Some patches, like `13_MorphLd` and `23_TiltPad`, have their own tilt movements built in. `32_Drone` falls apart as you tilt it.



What the tilt moves is set by `XTarget` / `YTarget` (destination) and `XDepth` / `YDepth` (depth) in `SET > IMU`.

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6. Add and remove FX

MORPHASE has five effects (Chorus / Delay / Reverb / Crush / Drive). While playing, these keys add or remove them temporarily.

Key	Effect	Letter on PLAY
`q`	Chorus (shimmer and width)	`C`
`w`	Delay (echo)	`D`
`e`	Reverb (room)	`R`
`r`	Crush (digital grit)	`X`
`t`	Drive (distortion)	`V`

Effects the patch uses are temporarily removed, and effects it doesn't use are temporarily added. This isn't saved; reloading the patch puts it back.

MORPHASE does not distort unless you use the distortion effects (Drive, Crush). When the sound gets too loud, the "output limiter" at the very end only turns the volume down, so nothing clips.

GLITCH (breaking it up on purpose)

While you hold `g`, the sound breaks up at random (repeating a fragment, playing backwards, chopping, knocking the tone off, and so on). Choose how in `SET > Glitch`.

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7. Change the sound a little

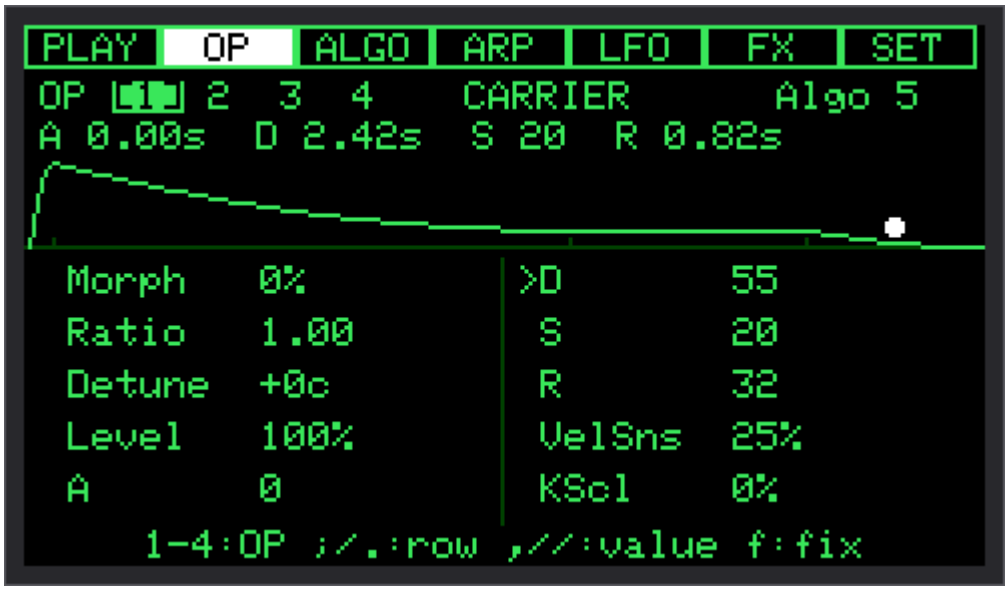
A MORPHASE sound is made of **four operators (OPs)**. An operator is a small oscillator that plays one waveform.

- **Carrier**: an operator that makes the sound you hear

- **Modulator**: an operator that wobbles a carrier's waveform to change its tone (you don't hear it directly)
- **Ratio**: the operator's frequency as a multiple of the note you play

Try it on the **OP tab**.

1. Go to the OP tab with `Tab`. Choose an operator with the number keys `1`–`4` (the screen shows `CARRIER` or `MODULATOR`)
2. Choose a setting with `;` `.` and change it with `,` `/` (hold to move in steps ten times larger)
3. Raising and lowering a modulator's **`Level`** changes the brightness a lot
4. Moving **`Morph`** changes that operator's waveform from a smooth sine towards sharper shapes
5. On the **`A`** **`D`** **`S`** **`R`** rows, an envelope graph (how the sound starts and fades) appears. Play a key and a white dot travels along the graph



However much you change, the factory patch stays as it was unless you save.

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8. Save a patch

Save from `SET > Patch`.

- **Save**: overwrite the current patch (asks to confirm)
- **SaveAs**: save under a name (up to 12 characters: letters, digits, `-` `_` `.`)

Factory patches (names starting with digits, like `01_`) **cannot be overwritten**. Choosing Save opens the screen for saving under another name (e.g. `01_EPiano` → `EPiano2`). If you delete or change them by mistake, `SET > Patch > Factory` restores them all.

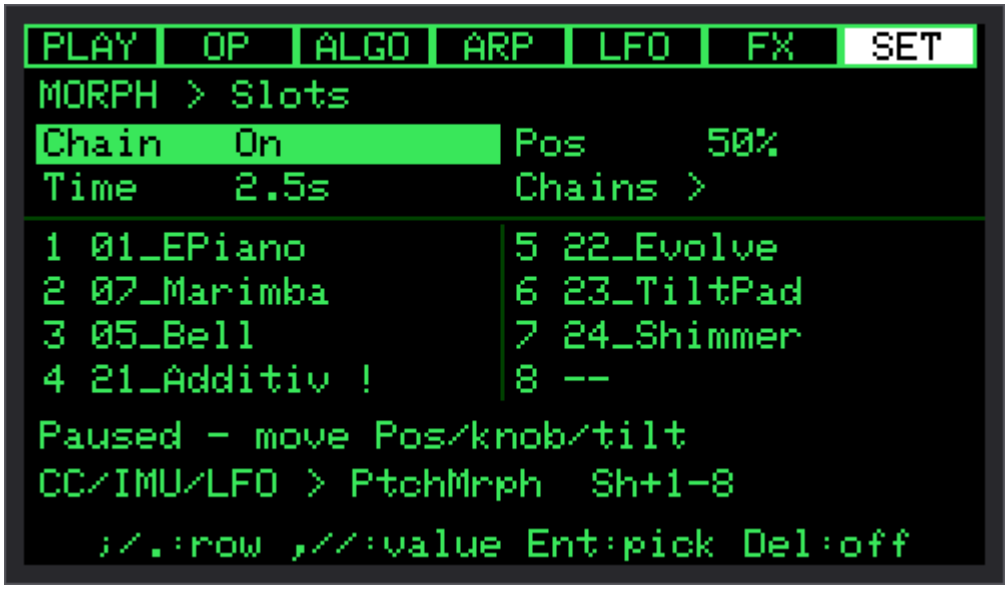
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9. Try the morph

The feature MORPHASE is named after. It moves the sound smoothly from patch to patch.

1. Open `SET > Patch > Morph` (titled `MORPH > Slots`). Eight slots hold patches (the "Pads" chain at first)

2. Press `Shift+1`–`Shift+8` to move from the current sound to that slot's patch over `Time` (1 second at first). The PLAY screen shows something like `M>3 45%`
3. To try drums, open `Chains >` and Load `02_Drums` (slot 1 = `25_Kick`, 2 = `26_Tom`)



The smoothest transitions are between patches with the same algorithm (the way the operators are connected). With different algorithms, the sound switches over part-way.

To switch from your MIDI keyboard's pads, set `SET > MIDI > PC Mode` to `Morph`; Program Change 0–7 then moves to slots 1–8.

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10. What MORPHASE can do

Here is the full picture, including the features you have already tried.

The FM engine

- **4-operator FM with 14 algorithms**: 8 built around the classic 4-operator FM layouts, plus 6 with feedback in different places (for sawtooth, growl and metallic sounds)
- **Waveform morphing**: each operator slides smoothly through 8 waveforms, from a sine to sharp shapes. `GMorph` moves all four together
- **Fixed-frequency operators and a pitch envelope**: build percussion such as kicks, toms and metal hits
- **12-note polyphony** (10 on the BLE build). Mono, legato and glide too

Patch morph: from sound to sound

- Moves between patches lined up in 8 slots, blending everything: ratios, envelopes, FX
- Two ways: a **chain**, where a knob, tilt or LFO moves the "position", and **glide**, which slides to a chosen patch with `Shift+1–8` or MIDI pads
- Chains can be saved under a name and switched one after another from a MIDI button

GLITCH: breaking it up on purpose

- Repeat the last fragment, play it backwards, chop it (Stutter / Reverse / Gate)
- Knock the tone off for a moment (Chaos)

- Break up the arpeggio rhythm (ratchets, octave jumps, running backwards, rests)
- Trigger it only while holding `g`, all the time, or as much as you tilt the unit

Theremin (M5Stack Unit ToF4M)

- **Wave your hand** over a distance sensor to move the sound
- Two modes: `Mod`, where hand distance moves the tone and more, and `Play`, where your hand's height plays a note like an instrument. In `Play`, pitch can glide smoothly or snap to one of 49 scales
- Play it along with the keys (chords with your left hand, theremin with your right)

Move it with your body

- **Tilt the unit** to change the tone (IMU; just turn it on and it works with most patches)
- Teach it 4 knobs and 4 buttons on a MIDI controller just by moving them (MIDI Learn)

Arpeggiator

- 5 patterns (Up / Down / UpDown / Random / AsPlayed) + **STEP**: 16 steps hold "which note of the held chord", so changing the chord keeps the rhythm and changes only the notes
- Latch, swing, sync to external MIDI clock, 8 saved patterns

FX, and an output that doesn't clip

- Crush, Chorus, Delay, Reverb, Drive. Add and remove them while playing with `q w e r t`
- An output limiter at the end: even big chords only get turned down in volume, never bent in shape

32 factory patches, ready to play

- From keys, bells, bass, leads and pads to drums and noise

Two builds

- Standard (Unit MIDI) and BLE, which also plays over Bluetooth MIDI from an iPhone / iPad (tested app: KeyPad)

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11. When you want to know more

The Reference Guide covers features such as:

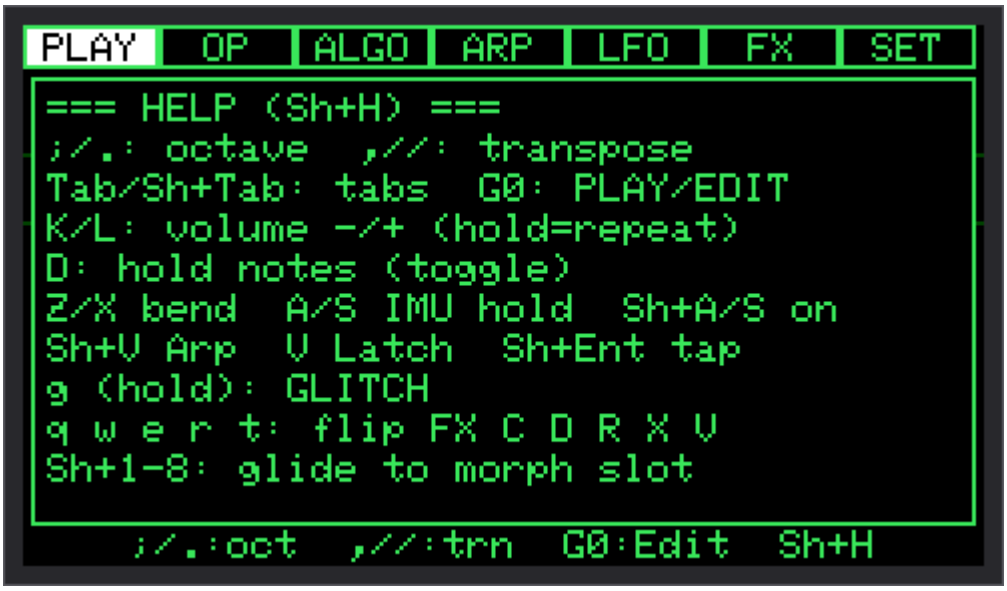
- The 14 algorithms, fixed-frequency operators, the pitch envelope
- The arpeggiator (including step patterns) and ARP glitch
- The LFO and the list of modulation destinations
- Assigning MIDI knobs and buttons (MIDI Learn)
- Saving chains and morphing in detail
- **The theremin** (plug in an M5Stack Unit ToF4M and move the tone or pitch with your hand)
- The list of 32 factory patches and what to try on each

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12. Troubleshooting

Start with the help

On any screen, `Shift+H` shows the keys for that screen. Press `Shift+H` again to hide it.



No sound

- Is the master volume (`VOL` at the bottom right of the PLAY screen) at 0? Raise it with `L`
- Is MIDI coming in? Play a key and see whether `MIDI` lights. If not, check the Unit MIDI's orientation, its DIP switch (BYPASS) and the Grove 5V switch (5VOUT)
- Does the receive channel (`SET > MIDI > Channel`) match your keyboard's send channel? (Omni at first)
- Is a microSD card inserted? (Without a card the initial sound plays, but nothing is saved)

The unit's keys stopped responding

Very rarely, for example after fast typing, the unit's keys can stop responding entirely. This is a known hardware quirk of the Cardputer ADV's keyboard chip (TCA8418). **MORPHASE detects it and restarts itself after about 30 seconds.** Just wait.

The BLE build won't connect

- Are you connecting from inside the app? (MIDI does not connect from the phone's Bluetooth settings screen)
- If you have registered the same Cardputer as a Bluetooth device with other firmware before, remove that old entry in the phone's Bluetooth settings
- Is `SET > MIDI > BLE` set to `On`? (Changing it needs a restart)

Still stuck?

Open a GitHub Issue or contact the developer (@Tokagetchi).

MORPHASE is an FM synthesizer made by an individual developer. Have fun moving the sound.